

The Country Space

Construction, Variable Choice, and Multi-Year Replication

Institutional Complexity Project

2026-05-01

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Overview

This document is the clean writeup of how we construct a 2D **country space** from V-Dem governance indicators. It pulls together the version of the pipeline we have settled on after experimenting with several alternatives in `institutional_complexity.Rmd` (the “kitchen”) and applies it to six benchmark years from **1975 to 2025**.

The pipeline in one sentence: take every continuous V-Dem indicator with adequate coverage in the target year, z-score across countries so each indicator contributes equally, and embed the resulting country x feature matrix in 2D with UMAP using Euclidean distance.

We deliberately avoid hand-picking indicators. The motivation, the consequences, and a few sensitivity checks live in Section 3.

1. Why a country space?

The Atlas of Economic Complexity uses a *product space* — a 2D map of products, with countries positioned by what they produce — to study how productive capabilities cluster and how countries diversify. The institutional analogue swaps “products” for V-Dem’s expert-coded measures of governance: rule of law, electoral integrity, judicial independence, civil-society autonomy, and so on. A country’s vector across these measures is its **institutional fingerprint**, and the country space is the 2D projection of those fingerprints.

The point is not the absolute coordinates (any 2D embedding of a high-dimensional space is heuristic). The point is that countries with similar institutional architectures end up close together, and the resulting layout exposes regional and regime-type structure that is otherwise hidden in tables of correlation matrices.

2. The data: V-Dem, year by year

V-Dem provides expert-coded scores for ~500 governance variables per country per year, with full back-coverage to 1789 for established polities. Most of the variables we use are produced by a Bayesian IRT measurement model that pools expert codings into a continuous latent score (typically in roughly $[-4, +4]$). This continuity is what makes Euclidean distance a sensible thing to compute.

We load the parquet snapshot maintained at `data/shared-data/vdem/vdem.parquet` and the companion codebook. Coverage by benchmark year:

Table 1: Country (polity) coverage by benchmark year

year	n_countries
1975	158
1985	157
1995	175
2005	177
2015	179
2025	179

V-Dem’s polity count grows over the late 20th century as decolonization and the Soviet collapse fragment large polities into many smaller ones. We treat each year’s panel as its own population: indicator selection and country selection are both done within the target year (see Section 3).

3. Variable choice

This is the most consequential design decision in the whole pipeline. Two reasonable positions exist:

1. **Hand-pick** a curated set of, say, 20-40 governance indicators, chosen on conceptual grounds. This is what most institutional-economics papers do.
2. **Take all of them** and let the metric do the work. This is what we do.

We pick (2) on the grounds that any hand-curated subset embeds the analyst’s prior about what governance “is” — judiciary vs. media vs. elections — and a country space is supposed to be a *prior-light* map. The cost is that V-Dem ships indicator families that triangulate the same latent concept (multiple measures of judicial independence, multiple measures of media freedom). With min-max + Euclidean, those families implicitly upweight the dimensions they cover. We accept this for now and flag it at the end of this section. The Mahalanobis-whitened variant in `institutional_complexity.Rmd` is the robustness check.

What “continuous indicators” means concretely

V-Dem’s codebook tags every variable with a `vartype`:

- **C** — Component: an expert-coded measurement, usually delivered as the IRT latent score. These are the raw signals.
- **A** — Aggregate / **D** — Derived: indices built from components (e.g., the liberal-democracy index). Including these would double-count the components they are built from.

We keep only `vartype == "C"`. From those we further filter:

- **Drop suffix variants.** Each base indicator ships with `_codelow`, `_codehigh`, `_sd`, `_osp`, `_ord`, `_mean`, `_nr` companions (uncertainty bounds, ordinal recodings, raw means, response counts). Only the base IRT score is kept.
- **Drop multi-coder per-rater columns** (`_1`, `_2`, ...). These are individual rater’s codings, not aggregated scores.
- **Drop two discrete-categorical components:** `v2regopploc` (“regime opposition location”) and `v2regsuploc` (“regime support location”) are integer-coded categorical variables, not IRT scores. Treating them as continuous numbers is meaningless and they would dominate Euclidean distance because their nominal-integer scale is much wider than the IRT $[-4, +4]$ range.
- **Coverage filter.** Within the target year, an indicator must be observed for at least 50% of countries. Rare or near-empty columns add noise without signal.
- **Drop a small set of near-duplicate indicators** (see below).

The filter is stated as a rule, not a list of approved tags. Applied to a different year, it will produce a slightly different indicator set as V-Dem coverage evolves (some indicators were added in later codebook waves; some have spotty coverage in the 1970s).

Pruning near-duplicate indicators

The “all C-type indicators” rule keeps several indicator pairs/triples that V-Dem ships as conceptual duplicates — most obviously, every civil-liberty indicator coded separately for men and women, plus a few cause/effect pairs (high vs. lower court independence; general CSO repression vs. religious-organization repression). Empirically these pairs sit at $|r| \geq 0.85$ across the 2025 cross-section and contribute almost nothing beyond the base indicator. Concretely, hierarchical clustering on $1 - |r|$ at a 0.15 cut surfaces 24

redundancy clusters covering 61 of the 192 indicators (see `outputs/redundancy_clusters_2025.csv` for the full table).

We do **not** prune all 24 — many of those clusters bundle conceptually distinct mechanisms that just happen to co-vary in 2025 (executive-corruption vs. public-sector-corruption; censorship effort vs. self-censorship; polarization vs. political violence). Collapsing those would lose interpretable signal in panel or counterfactual analyses.

We do prune a conservative list of **18 indicators** where the redundancy is structural rather than coincidental — gender-paired siblings, court-level pairs, religious-org- specific copies of general-CSO measures, and a handful of within-family duplicates. The v-dem advisor confirmed each as low-information beyond its kept counterpart. The full list is in the `redundant_drops` vector in the next chunk.

The motivation is *not* dimensionality reduction for its own sake (a 174-vs-192-feature UMAP is essentially indistinguishable). The motivation is interpretability: when a country sits unusually far from its neighbors, we want that to mean something substantively distinct, not “scored low on women’s property rights AND men’s property rights AND access to justice for women AND...” with each gender-paired indicator counting again toward distance.

The construction function

4. Building the matrix

We illustrate on the most recent year (2025) and then run all six.

```
## Year 2025

## Countries: 179

## Indicators: 173

## Mean per-feature SD after z-scoring (should be 1): 1
```

The matrix is 179 countries x 173 indicators. Each row is a country’s institutional fingerprint, each column is z-scored across countries so a one-unit move in any column equals one cross-country standard deviation in that indicator. Pairwise Euclidean distances on `Mz` are then directly interpretable as “total cross-country-standardized institutional difference” — large values mean two countries differ by many SDs across many indicators.

Why z-score, not min-max?

The earlier exploration in `institutional_complexity.Rmd` used min-max normalization to rescale every indicator to $[0,1]$. That works for the EC-style proximity calculations but distorts Euclidean geometry: a feature with a few extreme tail values gets squashed into the same $[0,1]$ range as a feature with broad variation, so the former contributes much less to inter-country distance than its actual signal warrants. Z-scoring puts every indicator on the same SD scale and is the right choice when the downstream operation is Euclidean distance.

Why mean-impute, not drop?

Coverage at the indicator level is already filtered ($\geq 50\%$ of countries). At the country level we drop rows with $> 50\%$ missing on the kept indicators. The remaining NAs (typically a few percent) are imputed at the column mean. This biases the imputed country toward the global average on the missing dimension, which is the most defensible neutral fill — it neither helps nor hurts the country’s distance to its true neighbors *on that dimension*, just shrinks variance toward zero on the imputed slots.

5. Embedding: UMAP with Euclidean distance

UMAP defaults to Euclidean distance, and we keep that default. We run on the full z-scored matrix ($\mathbf{Mz}$), not on a precomputed distance matrix or PCA truncation. That keeps the local-neighborhood structure UMAP is designed to preserve in terms of the metric we actually care about. We use mild hyperparameters (`n_neighbors = 15`, `min_dist = 0.2`) — small enough to surface clusters, large enough that the layout is not stringy.

Two caveats worth keeping in mind:

- UMAP layouts are stochastic. We set `set.seed(42)` for reproducibility, but a different seed (or a different indicator set on a re-run) can rotate, flip, or modestly rearrange the layout. Cluster structure is stable; absolute coordinates are not.
- 2D coordinates across years are **not aligned**. Each year is embedded independently. Compare structure (which clusters exist, who is in them, how isolated they are), not coordinates.

Country space, 2025



6. Multi-year replication

We run the same pipeline for the six benchmark years and lay them out as small multiples. The indicator set is re-derived each year from the coverage rule, so older years use a slightly leaner set (some V-Dem indicators were added in later codebook waves; others have thinner pre-1990 coverage).

Table 2: Pipeline output by year (countries kept; indicators kept).

year	n_countries	n_indicators
1975	158	188
1985	157	188
1995	175	190
2005	177	222
2015	179	222
2025	179	173

How indicator availability changes over the panel:

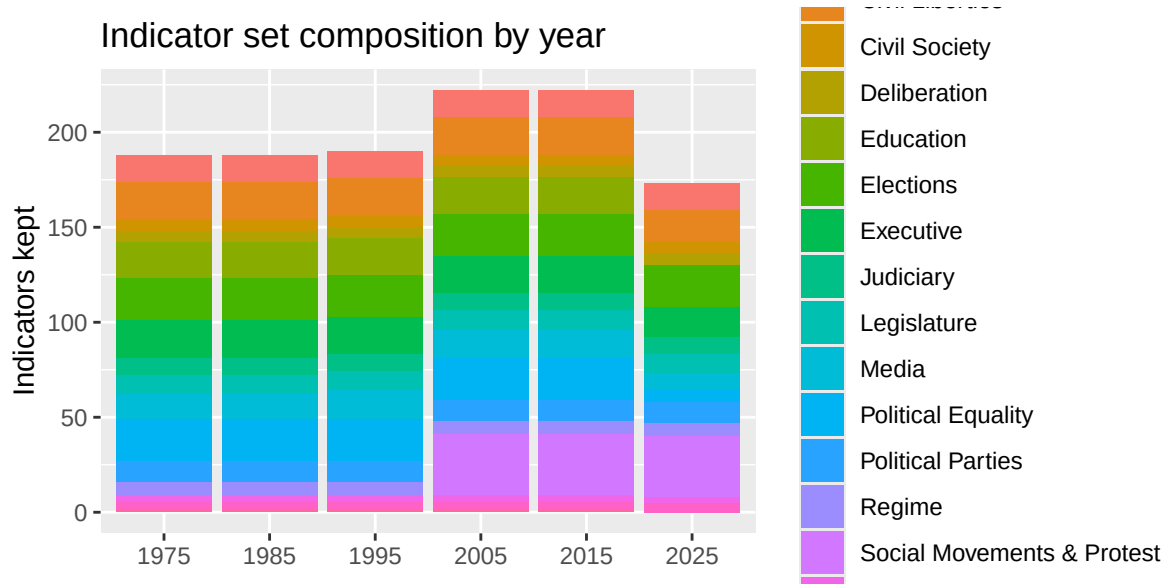


Table 3: Indicator-set overlap across the six years

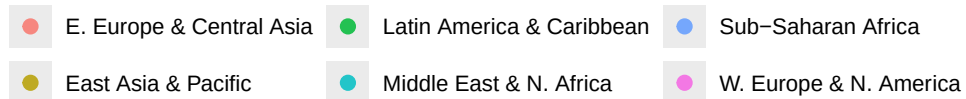
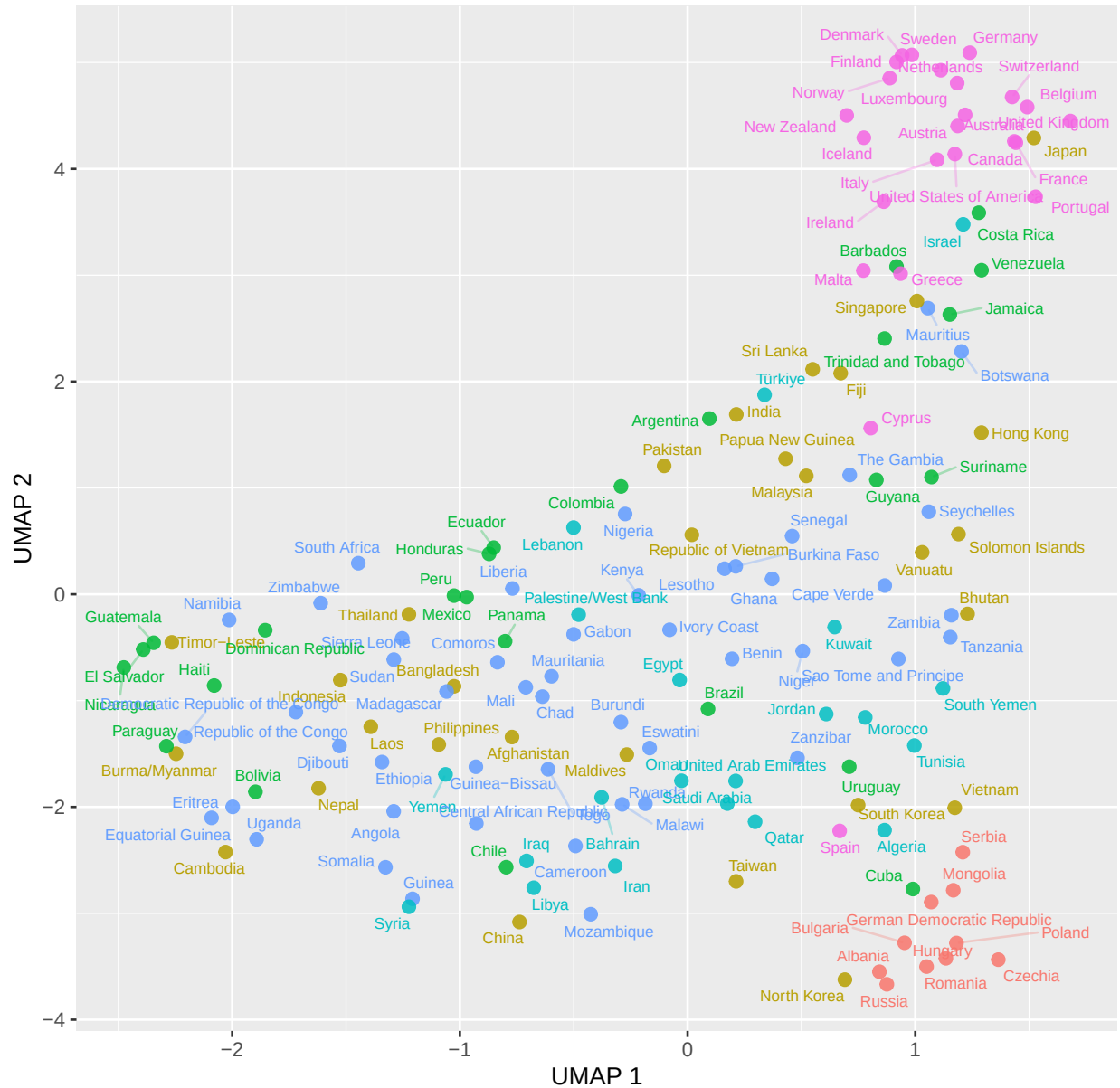
Metric	Value
Indicators kept in every year (core)	139
Indicators kept in at least one year (union)	222
Core / union ratio	0.63

The core set — indicators that pass the coverage filter in every benchmark year — is what we’d use if we wanted **strict** cross-year comparability of the input matrix. We report it but do not enforce it on the layouts: re-running with the year-specific sets gives each panel the most informative space available for that year, which is what a domain reader probably wants. Each year’s space gets its own page below.

1975

Country space, 1975

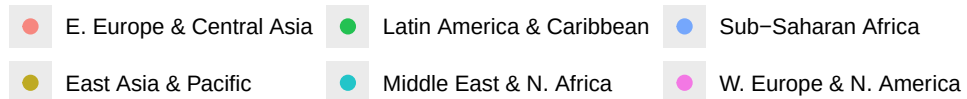
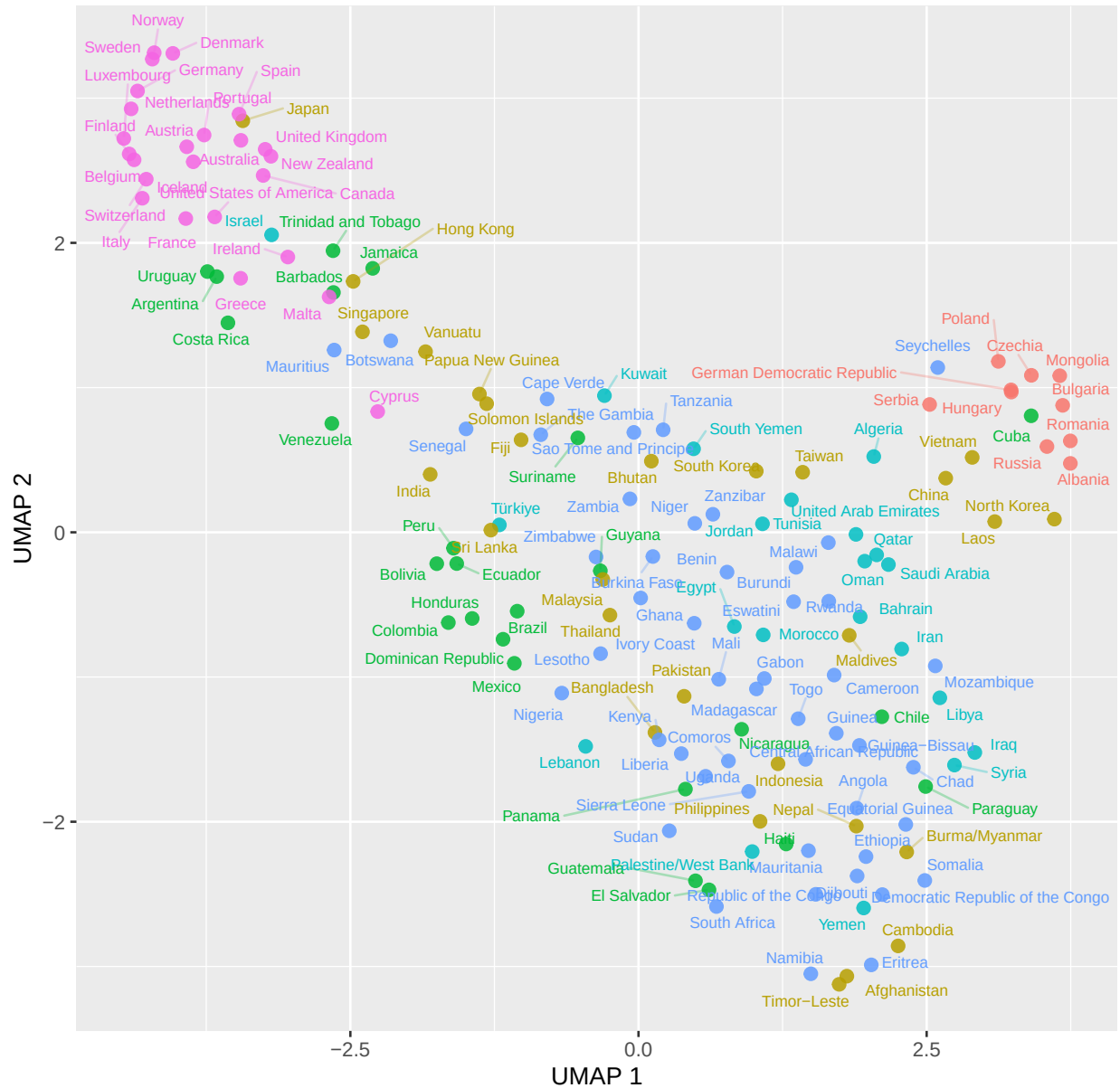
158 countries x 188 V-Dem indicators (z-scored, UMAP, Euclidean)



1985

Country space, 1985

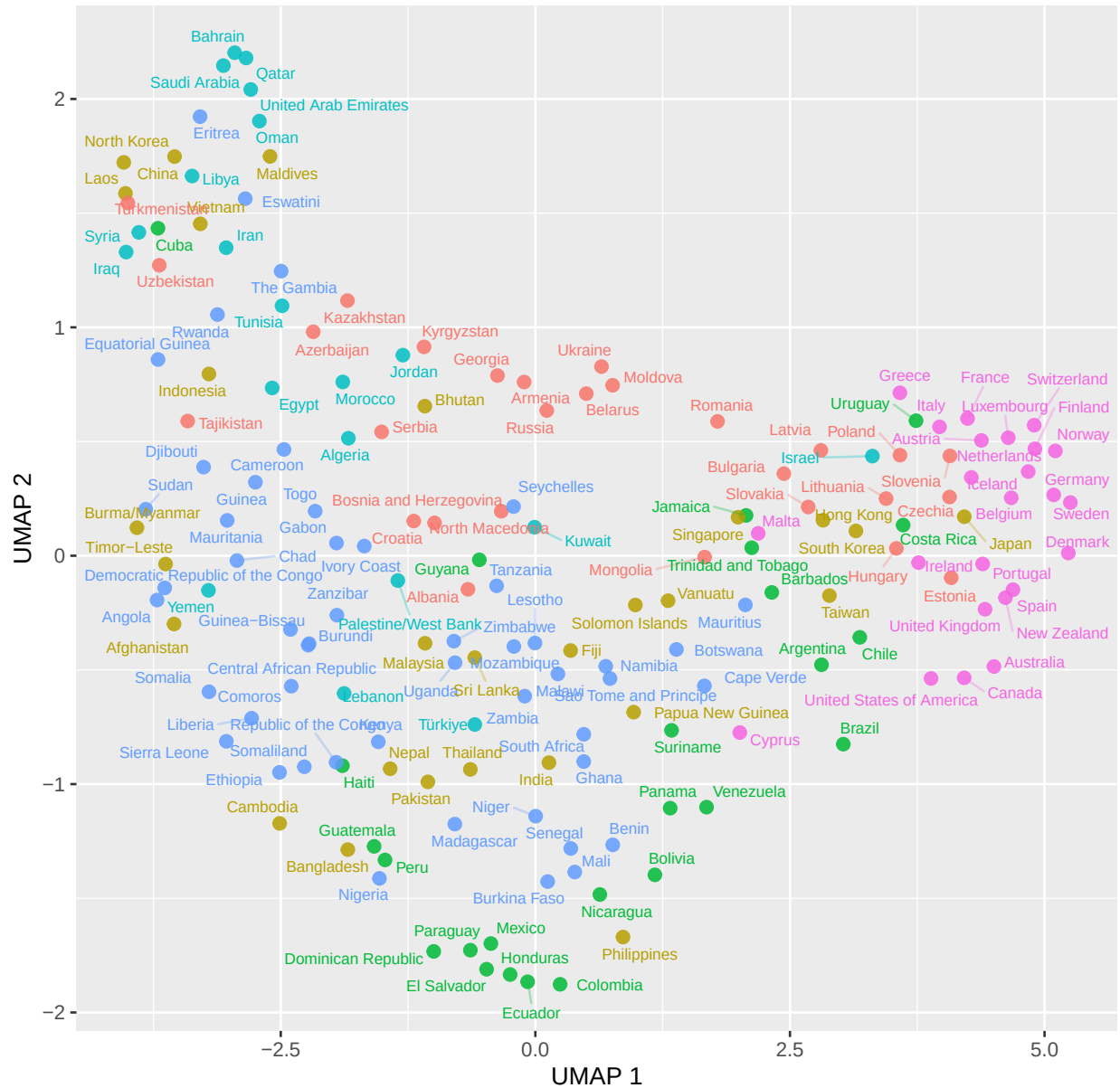
157 countries x 188 V-Dem indicators (z-scored, UMAP, Euclidean)



1995

Country space, 1995

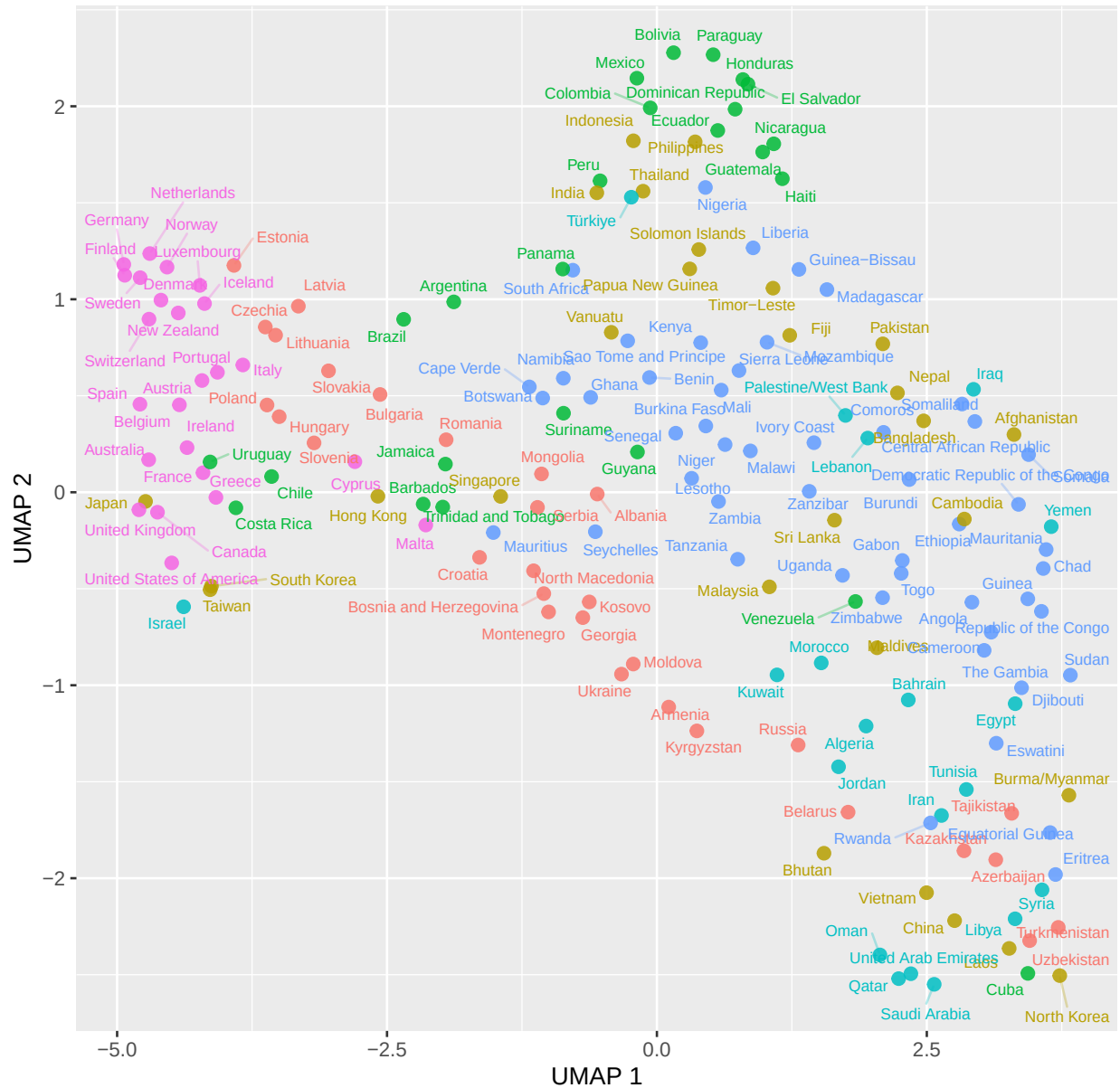
175 countries x 190 V-Dem indicators (z-scored, UMAP, Euclidean)



2005

Country space, 2005

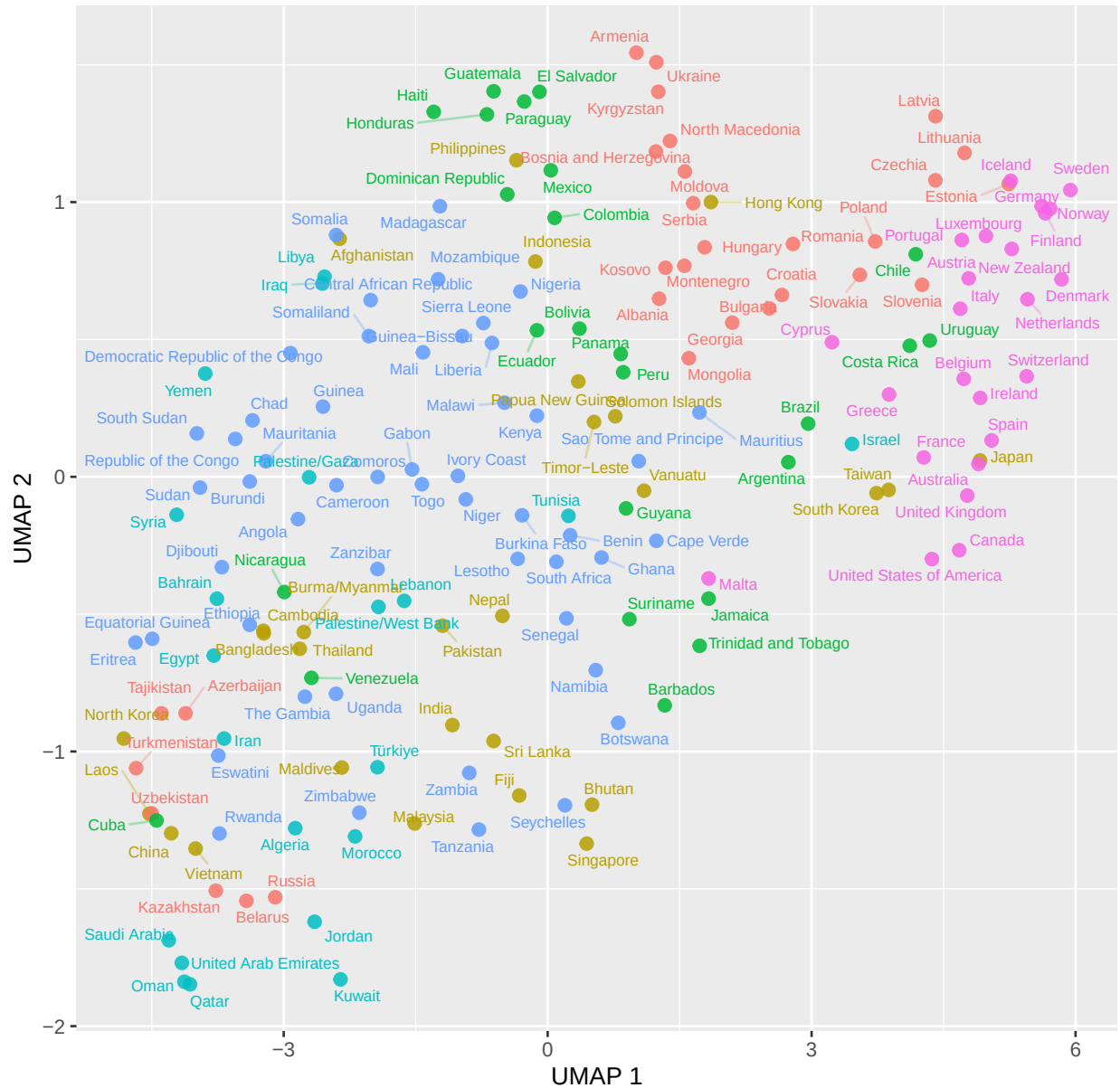
177 countries x 222 V-Dem indicators (z-scored, UMAP, Euclidean)



2015

Country space, 2015

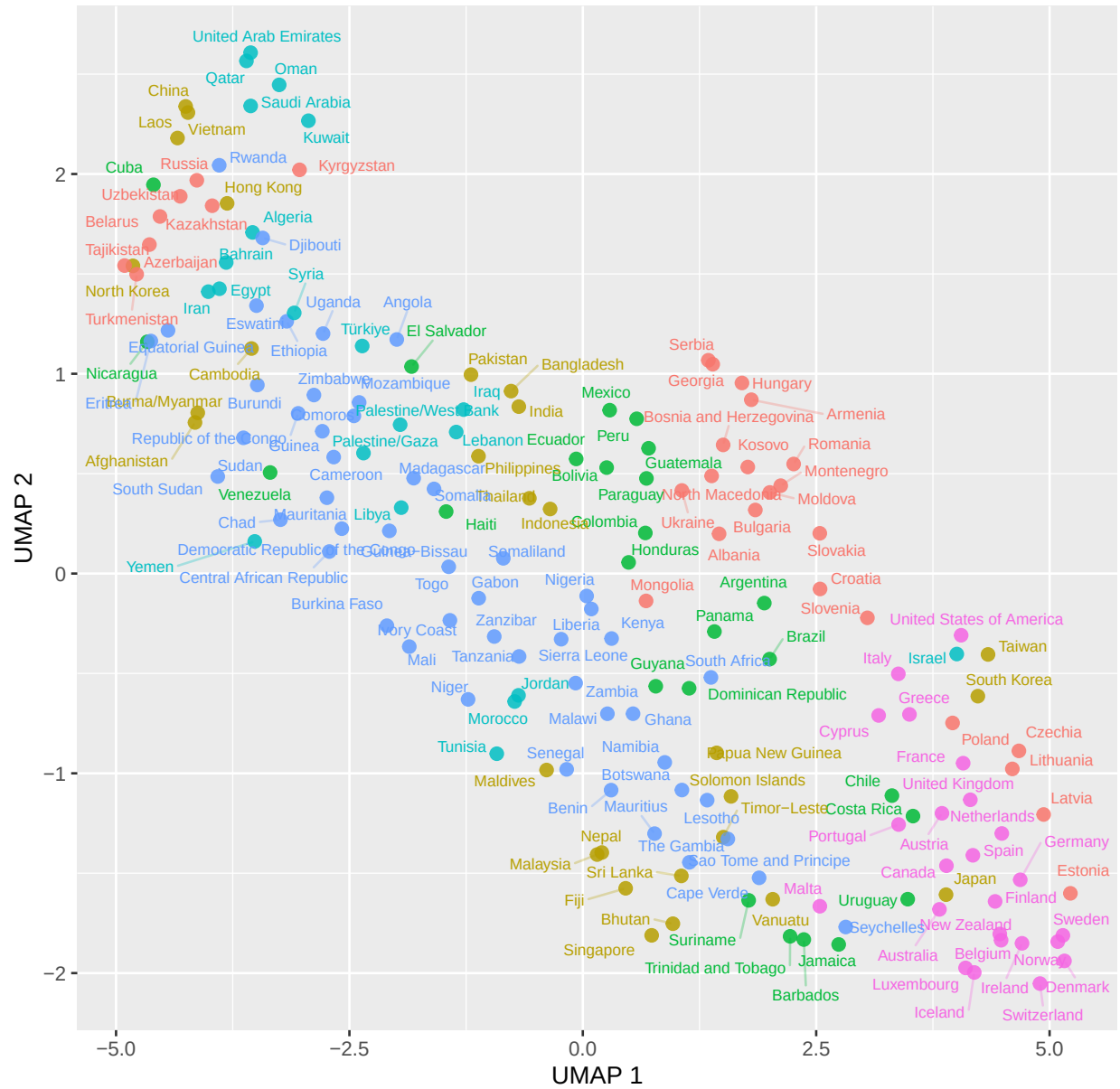
179 countries x 222 V-Dem indicators (z-scored, UMAP, Euclidean)



2025

Country space, 2025

179 countries x 173 V-Dem indicators (z-scored, UMAP, Euclidean)



7. What the panel shows

A few observations are worth flagging without over-interpreting:

- **Regional clustering is the dominant first-order pattern in every year.** That is reassuring: countries that share political-cultural neighborhoods do tend to share institutional features. It is also a bit of a Rorschach — the layout reflects genuine institutional similarity *and* the V-Dem coding tradition’s decision to benchmark some indicators against regional norms.
- **The Western Europe & North America cluster is tight in every year.** This is the “consolidated liberal democracies” anchor.
- **The post-Soviet space appears suddenly in 1995.** Pre-1991 the USSR is one polity; post-1991 it fragments into 15 successor states which then drift across the country space at different speeds (the Baltics moving toward the W. European cluster, Central Asia moving toward the closed-autocracy region, Russia carving out its own neighborhood).
- **The Middle East & North Africa region is unusually compact in earlier decades and starts to spread in 2015-2025.** This is the post-Arab-Spring institutional reshuffling visible in the data.
- **East Asia & Pacific is bimodal across the whole panel.** Japan/Korea/Taiwan/Australia pull toward the W. European cluster; China and its closer neighbors form a distinct pole. The middle (Indonesia, Philippines, India in adjacent regions) shifts year by year.

These observations are descriptive. The mapping from country-space neighborhoods to specific institutional features happens in `institutional_complexity.Rmd` (the cluster-feature z-deviation analysis at the end of Section 6 there) and is not reproduced here.

8. Limitations and what to read next

- **2025 is not the maximum-coverage year.** V-Dem’s indicator availability declines as we approach the present (242 indicators pass the 50% rule in 2020-2021, 215 in 2022-2023, 192 in 2024-2025). 2024 has identical coverage to 2025, so swapping to 2024 buys nothing; swapping to 2021 would gain ~50 indicators at the cost of four years of currency. We keep 2025 as the endpoint, but the alternative schedule (1971/81/91/01/11/21) is a one-line change to `benchmark_years` if richer coverage matters more than recency.
- **All-continuous-indicators** is a deliberate choice, not the only one. The Mahalanobis-whitened variant in `institutional_complexity.Rmd` checks how much the feature-family bundling distorts the Euclidean layout (answer: not much in 2025, but the comparison is worth re-running for older years before drawing strong cross-year conclusions about *structure*).
- **Mean imputation** biases imputed countries toward the global mean *only on the missing dimensions*. If a country has 30%-50% missing coverage, the imputation can pull it noticeably toward the centroid. For the most recent years this is rare; for 1975 and 1985 it is more common, especially among newly independent states. The per-country missingness distribution is a useful diagnostic to add if a specific country’s position seems off.
- **Coordinates are not comparable across years.** Treat each panel as its own map. A formal “trajectory” analysis — same country across years — would require either a joint embedding (concat the matrices, embed once) or a Procrustes-style alignment of per-year layouts. Both are reasonable next steps and are sketched in `case-studies/cambodia/cambodia-history.Rmd` for a single country, but not generalized.

- **Variable selection is the next thing to revisit.** The “all C-type indicators” rule is a starting point. A natural follow-up is to check whether dropping the top-loading indicator family (e.g., “all elections-related indicators”) materially changes the layout — that would tell us how much of the country-space is being driven by a single conceptual axis vs. genuinely multidimensional institutional variation.